



The trace of heat: on the predictive power of modeling transient diffusion

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Abstract

The paper at hand evaluates the validity of the transient heat equation with phase change and temperature-dependent coefficients as a model to predict the evolution of melt pools for rapid turnaround scan strategies in a laser powder bed fusion (PBF-LB) additive manufacturing (AM) process. In such scenarios, residual heat effects due to short laser-off times between tracks significantly impact the melt pool morphology. Results show that the conduction model accurately captures the rapidly evolving melt pool morphologies under these conditions. This is a fundamental discovery, as the ability to predict such dynamic scenarios has previously only been attributed to much more involved models which include fluid-like effects that otherwise require computational fluid dynamics (CFD). However, the presented validation (e.g., validated comparison to surface topography measurements) clearly demonstrates that a large portion of the observed effects is actually captured by the heat diffusion equation, including a common model of phase change. As a proof of concept for an application of this insight, feedforward control is investigated. Therein, the validated model predicts the melt pool morphology for a modified scan strategy with an increased pause time between the tracks that reduces melt pool distortions observed in the original configuration.

Keywords Thermal simulation · Model validation · Laser optimization · Turnaround effects

1 Introduction

Numerical simulations have the potential to reduce the time and cost associated with optimizing scanning strategies for laser powder bed fusion (PBF-LB) metal additive manufacturing [1]. A striking example of practical interest is the modulation of laser power to mitigate residual heat build-up and melting inconsistencies throughout a track, layer, or entire build. If melt pool morphologies and thermal evolution can be accurately predicted for a given scan strategy, these simulations can be used to directly optimize parameters according to a user's objective, for example, achieving faster build rates, mitigating systematic edge effects, residual stresses, and the like.

However, such optimizations require many transient simulations. They range from those based on simple models, such as (semi-) analytical models [2], and thermal simulations that consider only heat diffusion [3, 4], to advanced multi-physics models that also include evaporation and effects from fluid mechanics to accurately predict the temperature and melt pool evolution during the process [5–7].

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However, using these advanced models for transient optimization scenarios is nearly impossible due to their high demand for computational resources. For example, commercial software such as¹ Flow-3D required approximately 40 h to complete a 0.6 ms long transient simulation of a single 1 mm track on a moderately powerful workstation in 2015 [8]. Even though computational power has increased vastly in the last decade, detailed models, including melt pool dynamics, are still too costly for simulations on part scale. Thermal simulation models, on the other hand, which consider only heat diffusion and phase change, are simple and fast. Their speed makes them viable for process simulations up to the part-scale (see [9–11]). Such models also allow to accurately predict melt pool shapes and temperature distributions up to a certain “simulated maximum temperature,” regardless of the laser beam shape and scan speed [12]. However, model validations for melt pool morphologies have been carried out for quasi-stationary scenarios only, where the laser has moved sufficiently far to create time-invariant melt pool shapes. It was also demonstrated that this quasi-stationary (e.g., steady-state) setting allows the employment of heat diffusion models for laser beam shape optimizations to compute static beam profiles for various applications. Among these are the creation of laser beam profiles to generate rectangular-shaped melt tracks [13] or melt pools with uniform temperature distributions [14, 15]. Extending these thermal simulation models to transient scenarios allows for optimizing the speed of the laser, its power, as well as its shape or skywriting time. However, before those models can be used for such optimizations, it is necessary to explore their range of validity for highly transient scenarios.

In this contribution, the transient heat equation with phase changes is validated as a thermal simulation model to predict melt pool morphologies in non-steady scenarios. To this end, ex-situ measured transient melt pool morphologies created by two different scan conditions are compared to the model predictions. Section 2 includes fabrication details of the samples used in this study and implementational details of the thermal simulation. Section 3 compares simulation and experimental measurements of the melt pool morphology. An example of how the validated model can be used to evaluate simple feedforward control strategies is presented in Sect. 4. Section 5 summarizes the observations and takeaways from this study and outlines future work directions.

2 Methods

2.1 Artifact description and experimental setup

The PBF-LB artifact used in this investigation is depicted in Fig. 1 and referred to as a “rapid turnaround” (RTR) artifact [16, 17].

Its planar geometry consists of an isosceles trapezoid controlled by a minimum width w and maximum width W and included angle θ that is set to 5° . A total of 32 RTR geometries were fabricated (18 for the converging case and 18 for the diverging case) by incrementing/truncating the length of the sample (L_i) denoted at the top of Fig. 1. The geometry is fabricated onto a 76 mm $\times$ 76 mm $\times$ 6 mm IN625 plate using an ¹EOSM290 machine with vendor-recommended parameters for the alloy (hatch spacing $h = 110 \mu\text{m}$, power $P = 285 \text{ W}$, scanning velocity $V = 960 \text{ mm/s}$). The laser spot size ($D4\sigma$) is assumed to be $85 \mu\text{m} \pm 5 \mu\text{m}$. The skywriting option is turned on such that the laser is turned off after reaching the end of a track to enable the beam to decelerate and accelerate back to the programmed velocity before the laser is turned back on at the start of the next track (see dotted lines in Fig. 1). This allows for maintaining a constant linear energy density (e.g., $\text{LED} = P/V$) along the entire length of each track. The scanning strategy is configured such that for the same artifact geometry, a “diverging artifact” represents the fabrication case where each sequentially scanned track is longer than its predecessor, as opposed to the converging artifact, where each sequentially scanned track becomes shorter (see Fig. 1). Note that for the diverging case, the first track is fused at the narrow end of the geometry (e.g., Track B), whereas for the converging case, Track B is the last track that is scanned.

Although in-situ thermography could provide an indirect evaluation of the transient dimensional characteristics of the melt pool (e.g., length, width, area) during multi-track scanning, these measurements are highly subject to environmental noise, limited spatial resolution, and they require an extensive calibration process of mapping gray levels from the intensity images to emissivity corrected surface temperature values. The calibrated images must subsequently be thresholded using the assumed solidification temperature of the material to obtain a continuous isotherm that is presumed to represent the melt pool boundary. Top surface imaging via ex-situ microscopy provides a more reliable ground truth reference of these melt pool characteristics at the point of solidification. However, measuring the dimensional characteristics of each individual melt track of a multi-track fabricated artifact with this approach is challenging, as each new track partially remelts the previous one. This is why partial CAD geometries were fabricated, such that transient melt pool morphologies could be evaluated

¹ Certain commercial entities, equipment, or materials may be identified in this document to describe an experimental procedure or concept adequately. Such identification is not intended to imply recommendation or endorsement by the National Institute of Standards and Technology, nor is it intended to imply that the entities, materials, or equipment are necessarily the best available for the purpose.

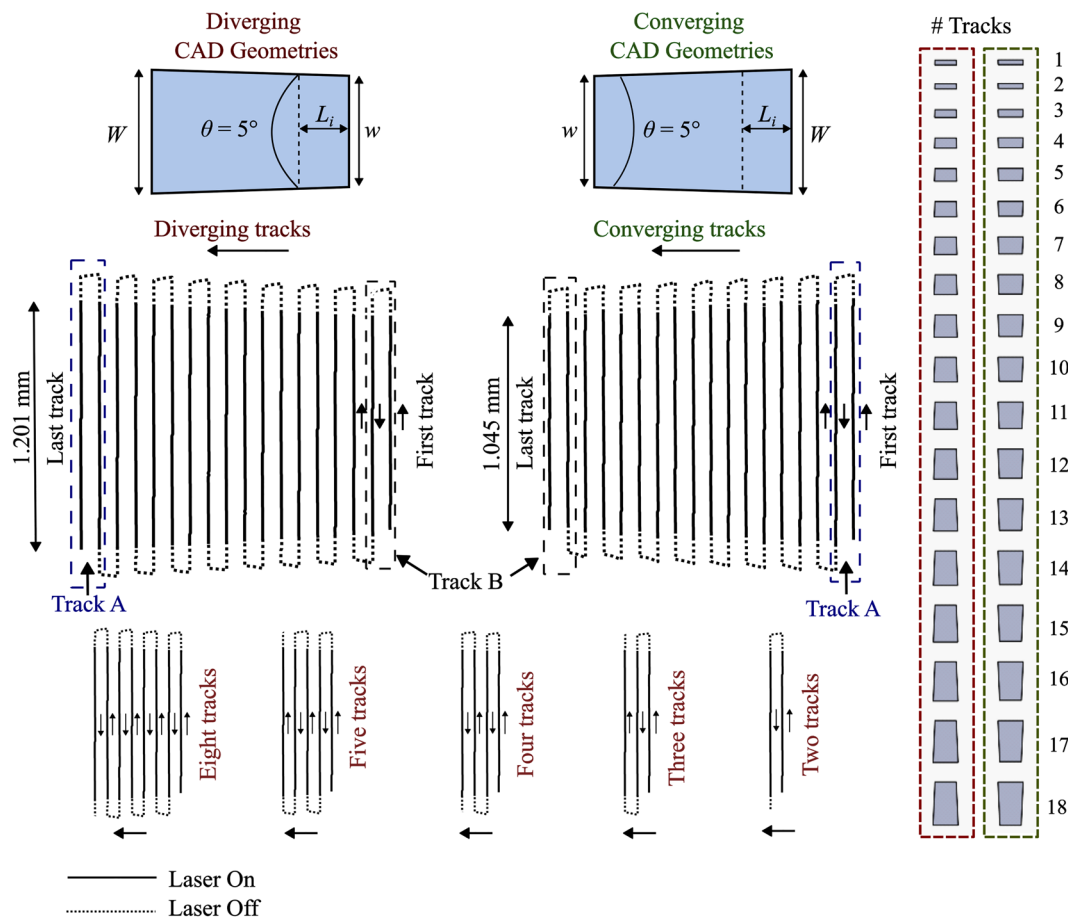


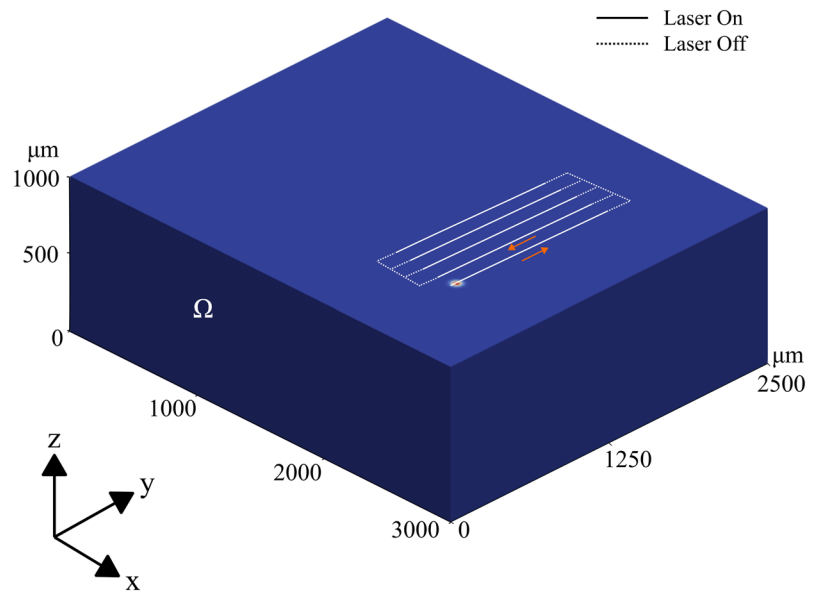
Fig. 1 Example of scanning strategies used in the fabrication of intermediate versions of a rapid turnaround (RTR) artifact geometry that are used to study transient melt pool dynamics

using ex-situ brightfield imaging of the top surface. Each partial geometry follows the entire scan strategy only up to a certain track number (1, 2, 3, 4,... up 18) and stops prematurely at that number to create solidified “snapshots” of the fabrication process (see bottom of Fig. 1). For the converging case, the first track starts at the wide end of the isosceles trapezoid, and for the diverging case the first track starts at the narrow end of the isosceles trapezoid (see Fig. 1). Two sets of converging and diverging geometries were scanned (set 1 and set 2) on different dates. In some cases, the vector lengths for an equivalent sample (e.g., 18 track) from set 1 and set 2 may vary ($\approx 150 \mu\text{m}$ or less) due to differences in the placement of the sliced geometries within the machine coordinate system. The top surfaces were imaged with a Zeiss AxioImager.Z2 microscope using ZenCore3.2 image acquisition and analysis software at 20 $\times$ (set 1) and 10 $\times$ (set 2) magnification, providing both spatial and intermediate temporal data for comparing simulated and actual solidified melt pool morphologies for each track of the fabricated artifact(s). Set 1 is used for qualitative analysis and model parameter calibration based on a single scan track.

Set 1 only includes diverging geometries with 2–8 and 18 tracks and a single converging geometry with 18 tracks. Set 2 contained three repeats of each converging and diverging geometry, ranging from 1 track to 18 tracks (see Fig. 1), and measurements from this set were combined with measurements from set 1 for quantitative validation of the model predictions for melt pool width and melt pool area. The melt pool width and area are measured after tracing the innermost chevron of the last track for each geometry in ImageJ based on brightfield images (See Fig. 7(a)). Note that tracing this inner chevron may involve some operator judgment. Hence, to consider this source of variation, all measurements from set 2 were acquired by two different operators, resulting in a total of at least 6 measurements (3 per operator) for each geometry shown in Fig. 1. However, in some cases, one of the three repeat samples exhibited noticeably different melt pool structures or significantly deviated from the nominally observed structures/trends of the repeated samples. This was likely caused by local contamination on the scanning lens, attenuating the laser power at the surface, and resulting in smaller melt pool sizes. Measured values in these cases

Table 1 Material and model parameters

Material and phase change parameters used for IN625			
Name	Property	Value	References
Density	ρ (kg/m ³)	8440	[21]
Solidus temperature	T_s (°C)	1290	[20]
Liquidus temperature	T_l (°C)	1350	[20]
Solid specific heat capacity	c_s (J/kg°C)	$410 + 276 \times 10^{-3} T$	[21]
Liquid specific heat capacity	c_l (J/kg°C)	766	assumed (c_s at T_s)
Solid thermal conductivity	k_s (W/m°C)	$9.8 + 15.3 \times 10^{-3} T$	[21]
Liquid thermal conductivity	k_l (W/m°C)	29.5	assumed (k_s at T_s)
Latent heat of fusion	L (J/kg)	2.8×10^5	[20]
Laser absorptivity	α	0.54	calibrated
Regularization	S	1	assumed

Fig. 2 Illustration of the domain and the scan strategy used for the simulations

were considered as outliers and removed from subsequent analysis.

2.1.1 Scan strategy alignment

The scan strategy itself is collected by a ¹Printrite 3D system by Sigma Additive, which samples the commands sent to the ¹EOSM290 controller at 100 kHz (e.g., every 0.01 ms) during part fabrication. This data is arranged according to the xy2-100 protocol. An example of the data in Hierarchical Data Format version 5 (HDF5) can be found in [18, 19]. The headers of interest are the x, y, laser power, and gating commands.

2.2 Model description

The following form of the non-linear transient heat equation is employed to model the temperature evolution during the laser melting on a bare metal plate:

$$\rho c \frac{\partial T}{\partial t} + \rho L \frac{\partial f_{pc}}{\partial t} - \nabla \cdot (k \nabla T) = 0 \quad \text{on } \Omega \times \mathcal{T} \quad (1)$$

$$k \nabla T \cdot n = \alpha u \quad \text{on } \Gamma_{N_L} \quad (2)$$

$$k \nabla T \cdot n = 0 \quad \text{on } \Gamma_{N_0} \quad (3)$$

$$T(\mathbf{x}, 0) = T_0 \quad \text{on } \Omega. \quad (4)$$

Therein, T and ∇T represent the temperature and its gradient inside the spatial domain Ω (as shown in Fig. 2) and a time interval $\mathcal{T} = [0, t_{end})$. The laser intensity, scaled with the absorptivity α is applied as a Neumann boundary condition on the top boundary Γ_{N_L} of the domain in Fig. 2. Furthermore, an adiabatic boundary condition (zero flux) is applied on all other boundaries Γ_{N_0} (left, right, front, back, and bottom boundaries), with $\Gamma_N = \Gamma_{N_L} \cup \Gamma_{N_0} = \partial\Omega \times (0, t_{end})$. The

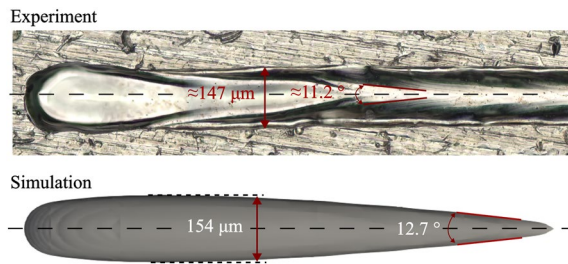


Fig. 3 Melt pool images from a single-track experiment and a single-track thermal simulation used for calibrating the absorptivity (α) in the model

initial condition of $T_0 = 20^\circ\text{C}$ is applied on Ω at $t = 0$. The coefficient ρ denotes the material density, which is assumed to be constant. By contrast, the specific heat capacity and the thermal conductivity linearly depend on temperature below melting (up to the solidus temperature T_s) and are assumed constant above melting. The model includes phase change through a phase change function $f_{pc}(T)$ (see e.g., [20]) that regularizes the transition between the solid (value zero) and the liquid state (value one) using the parameter S , with L denoting the latent heat of fusion. The assumed value $S = 1$ (i.e., no additional regularization of the phase-change transition) resulted in a chevron angle at the melt pool tail that approximately matches the single-track experiment result (see Fig. 3). Larger values of S resulted in a shortening of the melt pool length and an increase in the chevron angle at the tail of the melt pool. The material and phase change parameters for IN625 material used for simulations are depicted in Table 1.

Figure 1 is discretized using linear hexahedral finite elements with hierarchical refinements, as presented in [22]. The time domain is resolved by the classical Crank-Nicolson time integration scheme. The resulting non-linear equation system at each time step is solved using a Newton–Raphson scheme.

The simulations are conducted on a cuboid-shaped domain of size $3000\ \mu\text{m} \times 2500\ \mu\text{m} \times 1000\ \mu\text{m}$ as depicted in Fig. 2, which is approximately $500\ \mu\text{m}$ larger than the laser scan size to reduce the boundary influences on the results. The only unknown parameter in the model is the laser absorptivity, α . This parameter is calibrated to 0.54 based on the melt pool width measured from single-track experimental results as shown in Fig. 3. With the laser spot size ($D4\sigma$) of $85\ \mu\text{m}$, the simulation employs exactly the same laser scan strategies as used in the experiments. These include laser positions and laser power recorded at intervals of 0.01 ms. This is important for computing accurate results since the laser scan speed changes significantly during the

laser turnarounds where the laser is turned off due to the use of “skywriting”².

3 Results and discussion

The melt pool and its evolution are determined from the simulation by extracting the contour of the temperature field at the melting point ($T_s = 1290^\circ\text{C}$) at all discrete time steps. Figure 4 represents the thermal field within the simulation domain at the end of the 18-track diverging scan. The white line is the isothermal surface at $T = 1290^\circ\text{C}$ and represents the melt pool contour.

3.1 Qualitative validation

Figure 5 presents melt pool images from the experiment (on the left) and from the thermal simulation (on the right) for two, three, four, and seven incremental diverging scan tracks. The ex-situ melt pool measurements from the experiments (on the left in Fig. 5) show the final melt pool morphology in the last track and depict the evolution of the melt track through the chevron lines (as reported in [23]). The following effects occur:

- As the laser scans subsequent tracks, the melt pool width increases progressively (as observed from left to right in Fig. 5) to the point that the melt pool of the current track merges with the melt pool of the previous track which has not yet solidified (see the seventh track in Fig. 5).
- The major axis of the melt pool (see dashed line in Fig. 3) becomes increasingly misaligned with the programmed scanning trajectories (see e.g., the third track where the last melt pool appears tilted) until it merges with the previous track (see the seventh track in Fig. 5).

Both of these effects are matched by the computation using Equations (1) to (4). To highlight this, the melt pool morphologies are extracted from the computation by plotting the temperature traces at the melting point (melt lines) at discrete intervals of approximately 0.149 ms, until the end of the scan when the laser is turned off.

The predicted melt pool shapes are visibly similar in structure to the experiment results, with the melt track width gradually widening as the number of tracks increases. The simulation also predicts the misalignment of the melt pool

² Skywriting is a common feature used in PBF-LB scan strategies to ensure a constant energy density (e.g., P/V) is maintained throughout the scanning of each track. It involves turning off the laser at the end of each track to enable the beam to decelerate and then accelerate back to the programmed velocity before turning the laser back on at the start of the next track.

Fig. 4 Temperature field within the simulation domain at the end of the 18-track diverging scan, with the scan strategy approximately overlaid for reference

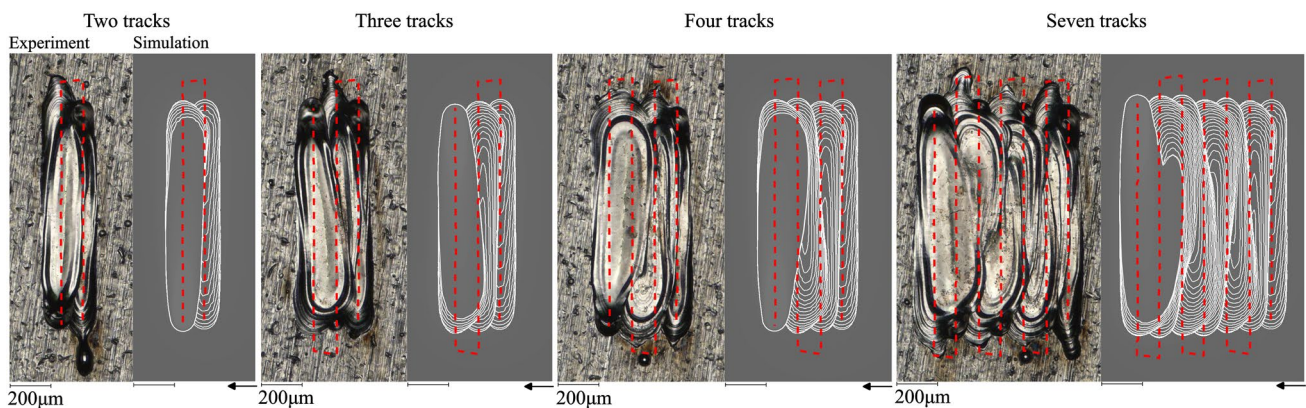
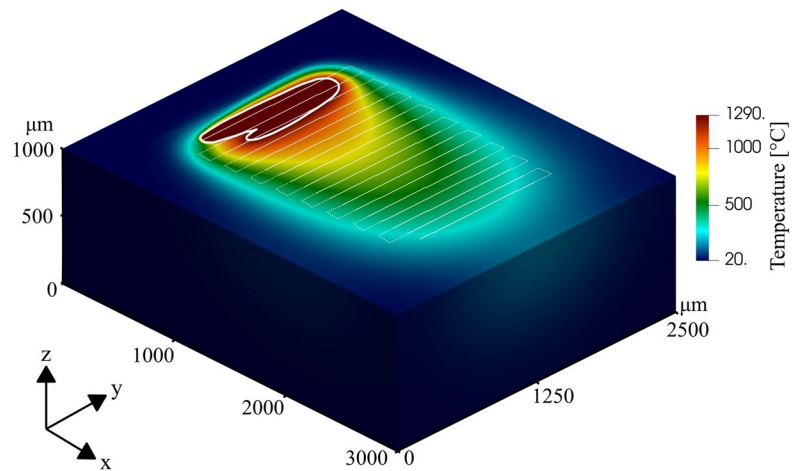


Fig. 5 Brightfield melt pool morphology images of the fabricated diverging scans for two, three, four, and seven tracks from the experiments (on the left of each image), and melt pool morphologies for

the corresponding scans from the simulations (on the right of each image). The red dashed lines represent the laser scan path. The black arrows below the melt pool images indicate the stepover direction

and its eventual merging with the previous tracks at the seventh track in Fig. 5.

These results are not by chance but consistently appear also for the 18-track scan for both configurations: (a) diverging and (b) converging vectors. Figure 6 presents the results from both the simulations and the experiments. Interestingly, there is a considerable difference in the final melt pool morphologies between the diverging and the converging scan strategies, even though the scans differ only by their successive hatch vector lengths. In the converging case, the last melt pool is wider and merges with two previous tracks, whereas in the diverging case, it merges with only one. In both of these cases, the simulation results are in excellent agreement with the experiments and capture these differences between the scan strategies accurately.

Both effects are caused by a residual heat build-up, which is inevitable in standard stripe-based (e.g., hatch-pattern) PBF-LB scan strategies. To quantify the residual heat build-up, the cumulative time above melt (CTAM) is computed

from the thermal simulation (shown in Fig. 6 on the right). The CTAM represents the total time that each spatial point spent above the melting temperature. This metric can be used to segment areas of the part that experience different thermal histories due to residual heat effects and phase transformation cycles, including remelting: (*powder* → *liquid* → *solid*) vs (*powder* → *liquid* → *solid* → *liquid* → *solid*). Figure 6 shows that the CTAM increases as the laser scans the tracks, both for the diverging and the converging scans, but more pronounced (approximately 7% more) in the converging case. The CTAM plots would be mirror-symmetric if there was no residual heat accumulation causing the effects listed above.

3.2 Quantitative validation

Figure 7 compares the evolution of the melt pool morphology in terms of dynamic maximum melt pool width and

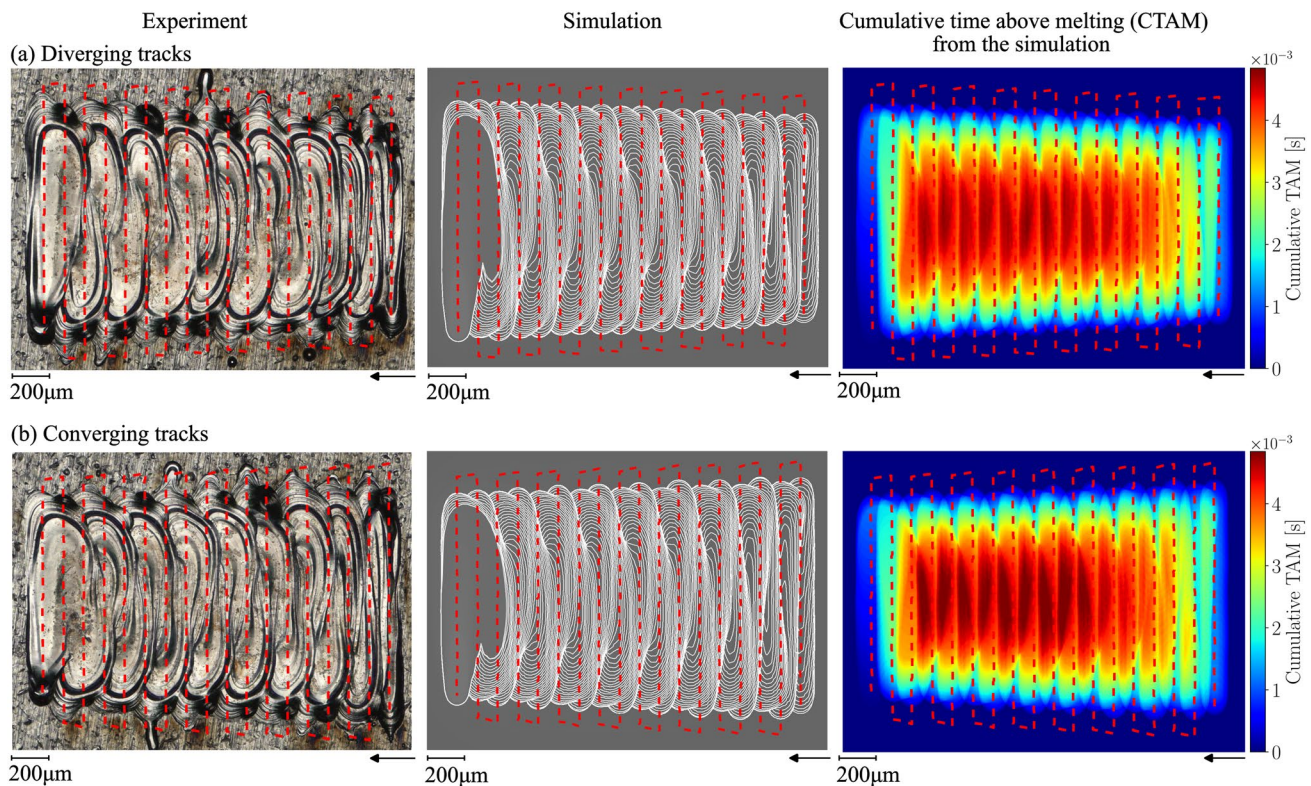


Fig. 6 Brightfield melt pool morphology images of the fabricated diverging and converging 18-track artifacts from the experiments (left), melt pool morphologies for the corresponding scans from the simulations (center), and cumulative time above melt (CTAM) heat-

maps based on the simulations (right). The red dashed lines represent the laser scan path. The black arrows below the melt pool images indicate the stepover direction

depth. Melt pool length values were not considered in this study as they are primarily influenced by the incremental changes in the programmed track lengths rather than residual heat effects between tracks. The experimental measurements of artifacts fabricated with 2 to 8 tracks and 18-track from set 1 (see Fig. 5 and Fig. 6), and 1 to 18 tracks from set 2 provide intermediate temporal validation of the simulation results for both the diverging and converging cases. They are determined by drawing a contour at an assumed melt line in the last track as indicated by the red line in Fig. 7(a). That line represents a “snapshot” of the instantaneous melt pool when the laser is turned off at the end of the scan. In-situ in-line laser illumination reported by Lane et al. [24] and Ricker et al. [23] have confirmed that the chevrons used to infer the assumed melt line indicate the edge of the melt pool.

The thermal simulation predicts the evolution of the melt pool width, depth, area, and volume, which are determined by the temperature computation. The simulation results show that the maximum width and depth of the melt pool increase with the number of tracks but saturate to a value in both diverging and converging cases. In the experiments, only the width and planar area can be measured (approximated as

shown in Fig. 7(a)). The results are plotted with “x” markers and vertical bars representing the average and standard deviation of the measured values for the width in Fig. 7(c) and area in Fig. 8(a). These measurements follow a similar trend for both the maximum melt pool width and the melt pool area. However, the experimentally measured melt pool widths are slightly larger than the predicted values, particularly after 8 tracks. Accounting for all the evaluated tracks for both converging and diverging cases results in an average difference in melt pool width of 14 % and a maximum deviation of 20 %. For the melt pool area, the experimental values differ on average by 11 %, with a maximum deviation of 22 %, except for the 1-track and 2-track scans, where the deviations reach up to 41 %. This is likely due to the slightly shorter scan length in experiment set 2 compared to set 1.

Comparing the plots between the converging and the diverging cases, Tracks A and Track B, shown in Fig. 7(b), have the same length in both cases, but significantly different melt pool widths (see Track A and Track B in Fig. 7(c)). This discrepancy is purely due to residual heat effects, as the power (P) as well as the scan velocity (V), and therefore the linear energy density ($LED = \frac{P}{V}$), are constant for each

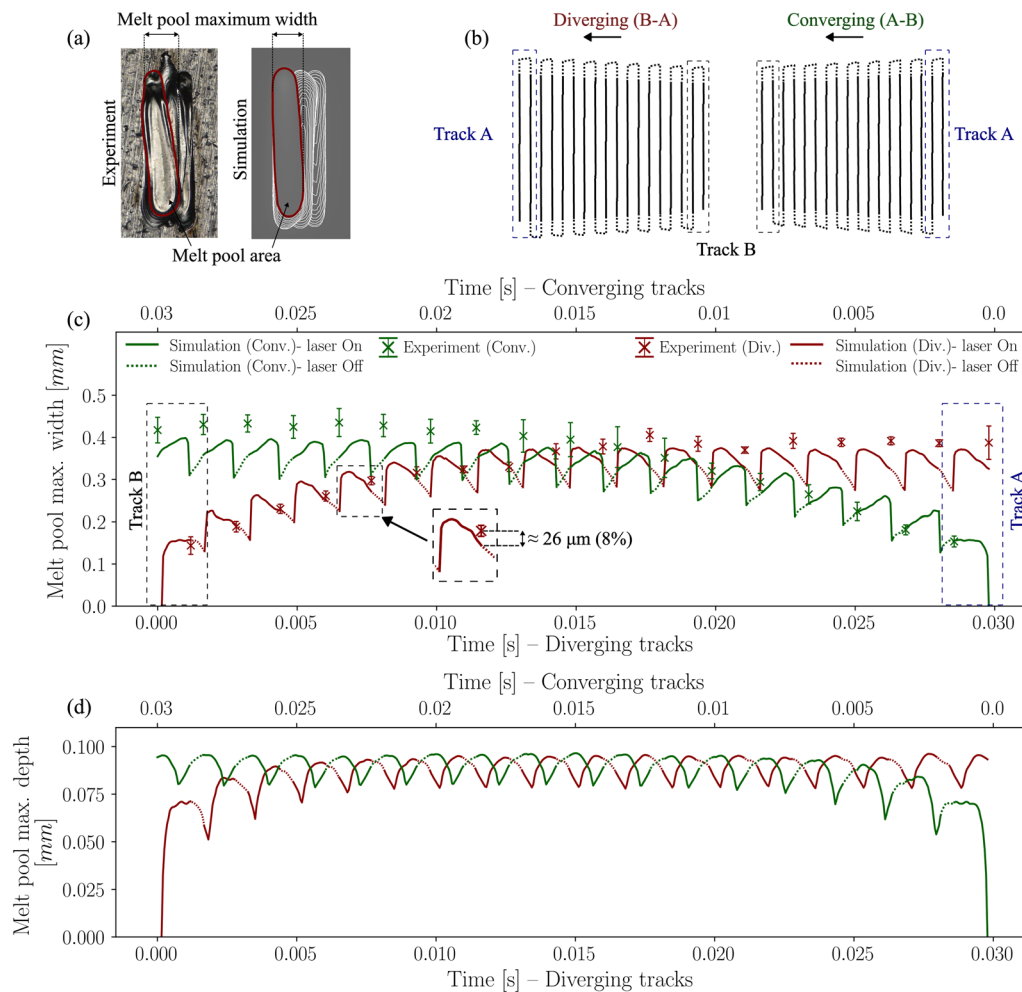


Fig. 7 **a** Example measurements of the melt pool area and width of the last track of a fabricated artifact. The red line indicates the assumed melt line. **b** Scan strategies (converging and diverging) for reference. **c** Comparison of instantaneous maximum melt pool width (c) and maximum melt pool depth (d) for the converging and diverging

cases shown in (b) from the simulation with the intermediate experimental measurements (marked by an “x” indicating the average, and a vertical bar representing the standard deviation) of the last track(s) overlaid. Equivalent track lengths for the diverging and converging cases are denoted as Track A and Track B

scan track. In the diverging case, Track A is the first track that is scanned, whereas, for the converging case, Track A is the last. Comparing the melt pool widths, Track A in the converging case (black dotted box) is increased by 2.5 times compared to Track A of the diverging artifact. On the other hand, Track B for the diverging case is about 2.4 times larger than Track B in the converging case. Although the experiments presented here were beam-on-plate, the trends regarding residual heat effects on melt pool depth matched those of high-fidelity multi-physics modeling results of melted IN625 powder presented in [25]. Those simulations showed that the residual heat from the previous track leads to the formation of a deeper melt pool at the beginning of the second track.

Figure 8 compares simulation results for the instantaneous melt pool area and melt pool volume. The estimated melt pool areas from the experimental images agree well with the

simulation results at the end of the tracks for the available scans from the experiment. Notice how the melt pool width in Fig. 7(a) approaches a maximum value at the point where the laser starts a new track. However, in Fig. 8(a), the melt pool area approaches a cyclic maximum near the end of a given track.

The dynamic melt pool evolution observed in the converging and diverging scanning strategies in the experiments may appear to be driven by fluid effects that drastically alter the melt pool morphology, seemingly necessitating advanced numerical models to predict the observed behavior. However, the results of these comparisons demonstrate that a thermal simulation model, which considers only heat conduction and phase change, is sufficiently accurate to predict melt pool evolution even in highly transient scenarios like those investigated here. This finding is significant, as it

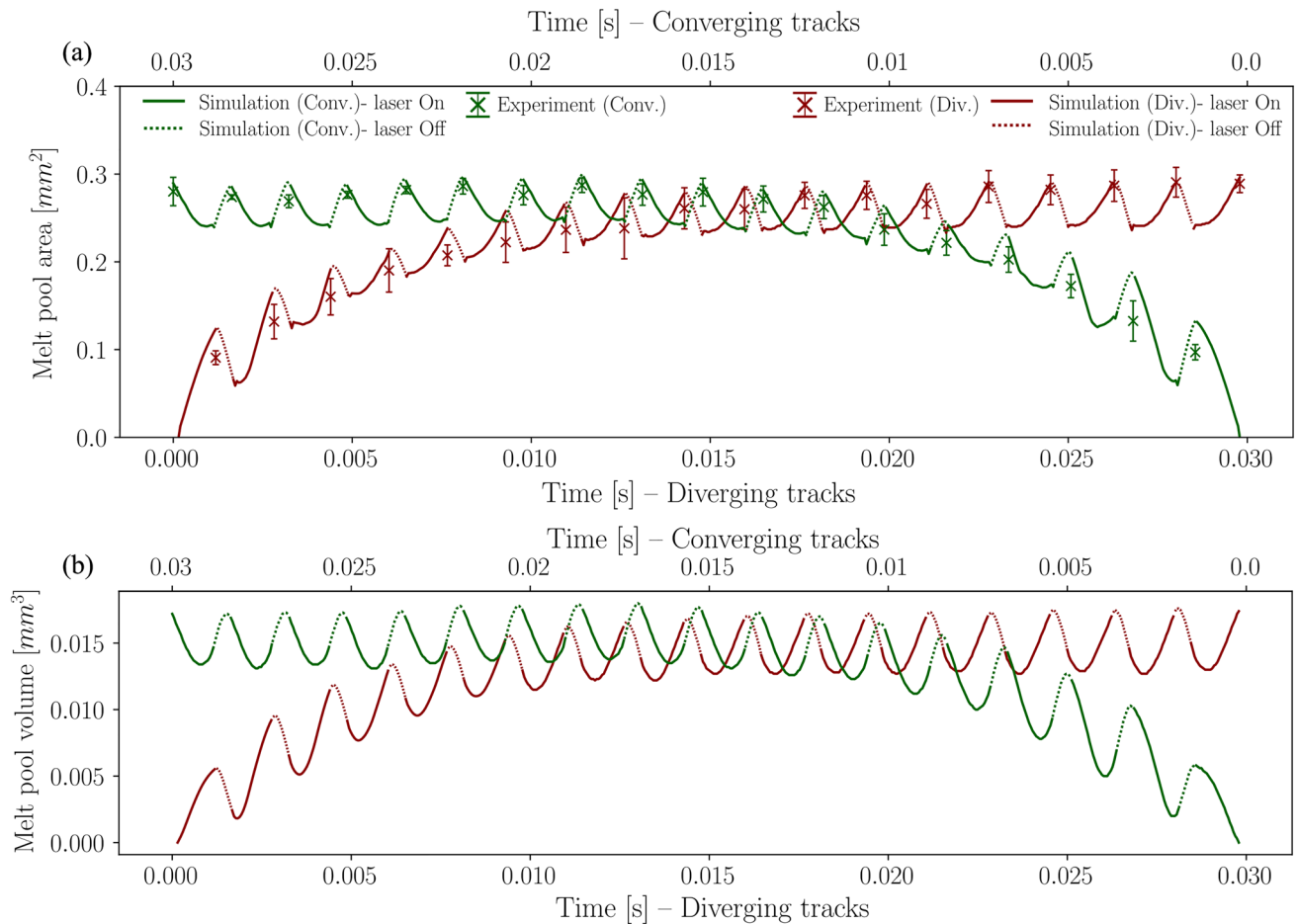


Fig. 8 **a** Instantaneous melt pool area vs time. **b** Instantaneous melt pool volume vs time

suggests that these simple and fast thermal simulation models can be reliably employed not only for part-scale simulations but also for numerical optimizations of laser scan strategies³. For instance, they can be used to optimize laser power profiles, scan speed profiles, or dynamic laser beam shapes, to mitigate these undesirable transient effects and to achieve user-defined objectives.

3.2.1 Remarks

Although no experimental results from this study are available for the melt pool depth and volume, the predicted average melt pool depth of approximately 92 μm (see Fig. 7 (d)) is comparable to the NIST AMB2022-03 benchmark melt pool depth measurements of $102.4 \mu\text{m} \pm 1.0 \mu\text{m}$. These measurements are based on fabricated single tracks for a comparable alloy (IN718 vs IN625) and Gaussian beam diameter, using the same laser power and speed

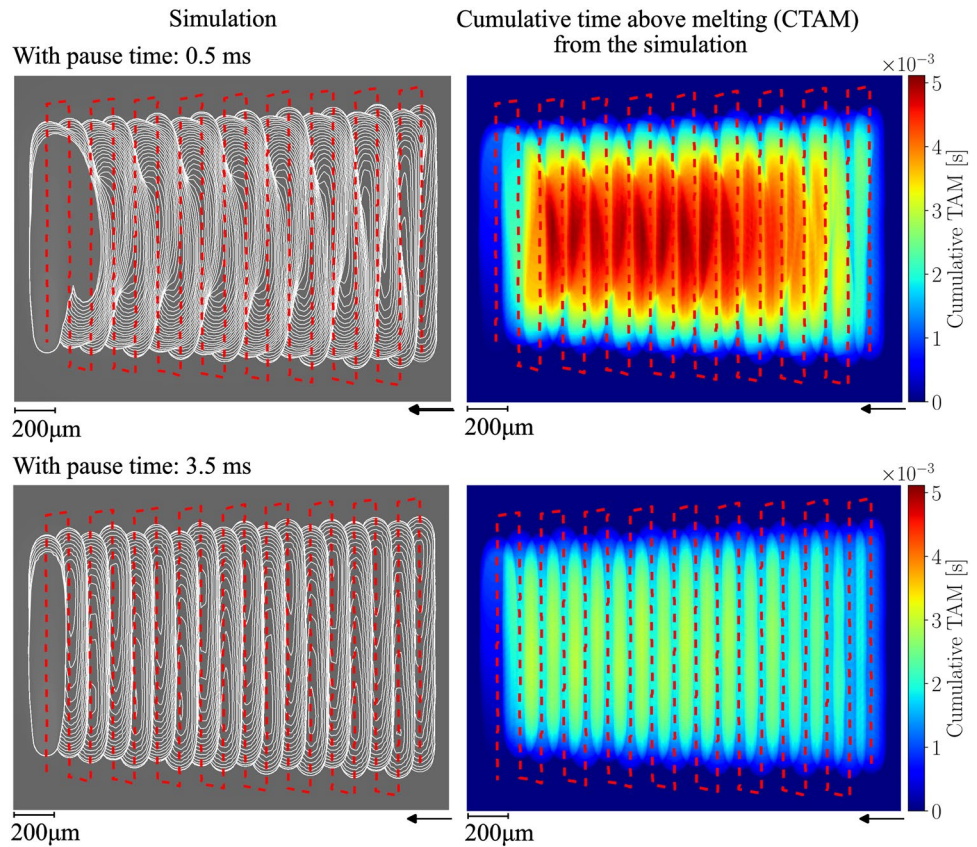
parameters and ambient conditions [26, 27]. This suggests that the thermal simulation model can reasonably predict melt pool depth, even if it slightly underestimates the values based on the parameters used in this work. However, this needs to be confirmed through thorough validation, as the model has previously been shown to underestimate melt pool depth—while predicting melt pool width with reasonable accuracy—for similar laser parameters applied to stainless steel 316 L, as reported in [12].

4 Considerations and examples of feedforward control

Conventional skywriting is usually employed to maintain a constant linear energy density ($\text{LED} = \frac{P}{v}$), enabling the laser to decelerate and accelerate back to a set scan speed before the fusion of the next track. However, depending on the machine's galvo system (e.g., mirrors that direct the laser scanning motion), the allocated time may not always be sufficient to enable the previous track to cool

³ The average computational time for one track is 14 min on an 2.6 GHz Intel Core i7-10750 H machine.

Fig. 9 Comparison of melt pool morphology and cumulative time above melt (CTAM) for the simulated ¹EOSM290 scanning strategy for IN625 and the adjusted scan strategy where the pause time at each turnaround is increased from 0.5 ms to 3.5 ms. The red dashed lines represent the laser scan path. The black arrows below the melt pool images indicate the stepover direction



adequately. In the context of PBF-LB, feedforward control proactively addresses thermal heterogeneity by taking corrective action before undesirable thermal conditions are created rather than reacting after the fact (e.g., feedback control). However, for feedforward control to be implemented effectively, a valid model is needed to predict how different scanning strategies (e.g., dynamic modulation of laser path, power, velocity, and shape) affect the local thermal conditions and melt pool morphology. Prior experimental work by [17, 28] developed a process-informed surface measurement technique. For the same material/build setting used in this work, they estimated a minimum of 3.5 ms between melt pools of adjacent tracks to be required to negate the residual heat effects, i.e., to avoid an increase in the melt pool width and misalignment of the melt track.

Figure 9 demonstrates the effectiveness of a simple feedforward control strategy to negate residual heat effects on the dynamic melt pool morphology using the validated thermal simulation model. The figure compares the melt pool morphology (on the left) and the cumulative time above melting (CTAM) (on the right) predicted from the simulation for the default ¹EOSM290 scanning strategy versus the modified case where the skywriting time (laser-off time) at each turnaround is increased to 3.5 ms.

Figure 10 compares the melt pool dimensions between the two scans. The results show that increasing the pause time reduces the residual heat effects on melt pool morphology with the subsequent tracks having the same width as the previous one, unlike the default case. However, this approach has the drawback of increasing the build time by approximately 166 %, as it takes 0.08 s instead of 0.03 s to fabricate the same sample geometry. The relative increase in build time with the added pause would have less of an effect in areas with much longer scan tracks. An optimal feedforward strategy should produce a consistent cyclic melt pool morphology through the scanning progression without slowing down the process. This could be achieved by varying either the laser power or the velocity or possibly adjusting the beam shape. All these parameters can be optimized numerically using the thermal simulation model within the machine's controllable limits.

5 Summary and conclusions

This contribution investigates the validity of the transient heat equation, including phase change, to predict the melt pool morphology in highly transient scenarios. Two rapid turnaround scan strategies with short, diverging, and converging hatch vectors are considered for the

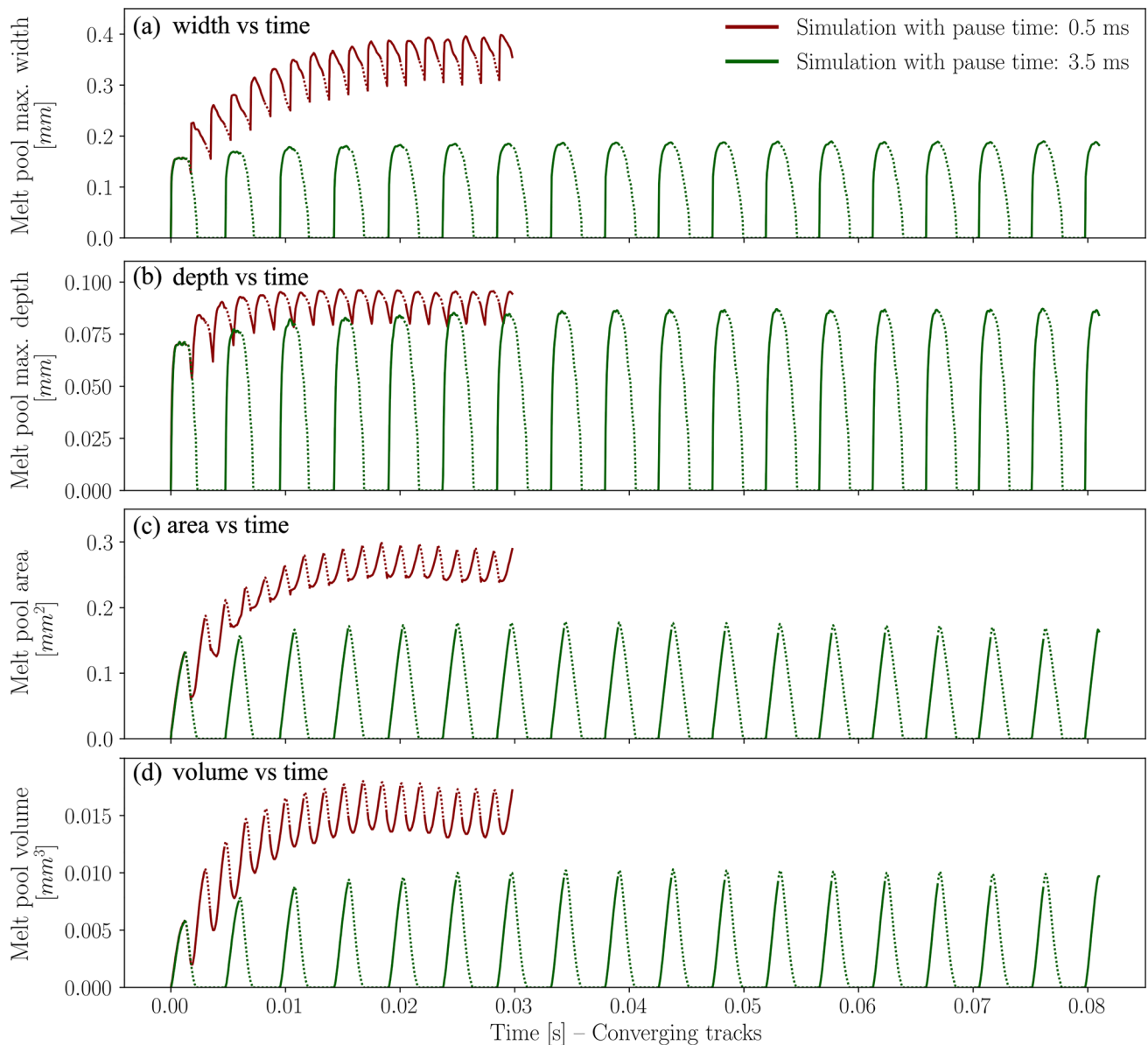


Fig. 10 Comparison of instantaneous melt pool width (a), depth (b), area (c), and volume (d) based on the simulated ¹EOSM290 scanning strategy for IN625 used for artifact fabrication and the adjusted scan

strategy where the pause time (i.e., skywriting time) at each turn-around is increased from 0.5 ms to 3.5 ms

validation. These scans result in residual heat build-up due to short laser-off times between tracks, leading to significant increases in the melt pool width and area along the xy-plane and track misalignment towards the previous scans. The results demonstrate that the thermal simulation model accurately predicts the melt pool morphology, even for dynamic scenarios as depicted in Figs. 5 and 6. This is a crucial insight as it has previously been thought that only more complicated models involving fluid dynamics could accurately predict the trace of the laser and the morphology of the melt pool for transient scenarios. The model was validated only where the melt

pool is in conduction mode in a quasi-stationary scenario. However, in this important case of practical relevance, the model can be used for transient numerical optimization scenarios such as optimizing laser power, scan strategy, and dynamic laser beam shapes. As an example, the converging scan strategy with an optimum pause time of 3.5 ms—determined using the method described in [17, 28], is simulated using the validated model. The simulation confirms that increasing the wait time to this value mitigates the residual heat effect, maintaining almost constant melt pool dimensions as the number of tracks increases. Future work will focus on further validating the thermal

simulation model for transient scenarios using in-situ temperature measurements with an infrared thermal camera (as shown in [24]) or multi-spectral camera ([29]). In addition, the predicted melt pool depth evolution will be validated using either cross-sectional imaging or synchrotron X-ray analysis measurements similar to [30]. The validated thermal simulation model will then be used to design optimal control strategies (using numerical optimizations) that dynamically adjust the process parameters to keep melt pool dimensions (width, depth, length, and area) within a desirable range by mitigating residual heat effects in such a manner that build efficiency is maximised.

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Data availability The simulation code and experimental data associated with this study are publicly available. The simulation code can be accessed at https://gitlab.com/hpfem/publications/2025_laser_turnaround_simulation_AM and the experimental data can be found at <https://doi.org/10.18434/mds2-3662>.

Declarations

Conflict of interest The authors declare no competing interests.

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